

ActInf Textbook Group ~ Cohort 3 ~ Meeting 20

(Chapter 9, part 1)

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Duration: 54:27 | Uploaded: Unknown Transcript method: auto_caption

all right hi everyone it's cohort three 725 23 and we're in our first discussion on chapter nine here so let us head over to chapter nine and begin before we go into any questions with just any General thoughts or comments that people want to bring up about chapter nine model based data analysis even just what what however in-depth people have read this like what what would they expect such a chapter to cover what would somebody want this type of a chapter to cover what is this title bring up for people or the quote from President Obama well I'll I'll just um give thoughts but but please write in the chat or raise your hand or or um you know feel free to to add in more great yes yeah thank you it's totally fine across different cohorts there's like not really a difference and just it's so many coats of paint that skipping around it's it's not um it's not how to line um and also in a sense chapter nine in specific I mean let's look at the textbook chapter one and chapter 10 are very similar that they're both overview chapters there's no equations they get big picture chapter two and three we know the low road chapter four technical details with generative model chapter five a lot of neurobiology and examples chapter six with a recipe for Designing active inference models probably makes more sense after knowing what the generative model is however probably you don't have to know it in technical detail chapter seven and eight on the discrete and continuous time generative models again probably helps to understand a generative model so seven and eight might rely on four but nine in a sense stands alone and so I hope that we can like dive into some of the intuitive and immediately accessible components of chapter nine because it's going to be a very different tack than the approaches from the previous chapters so let's get into it in this chapter we focus on the ways in which active inference can be applied in understanding empirical data now in all the previous sections the examples whether it's the frog jumping out of the hand or the musical notes or the rat in the Maze or the bird song all of these examples follow kind of a similar pattern which is like a generative model is specified and then it's played out it generates data that's one reason why these are called generative

models because these are models that generate data however in the real world we want to go both ways we want to be able to specify a model a priori and then generate data but also we want to have structured data and then do inference about hidden States for example given the empirical data so we don't just want to spin out mountains of data just to show that we can we want to take the data that we have which is called empirical and then do some kind of inference with it our general goal is to recover the parameters of the generative model that a subject's brain or mind uses to produce Behavior the subjective model so if all we were interested in was descriptions of behavior we wouldn't need sophisticated modeling at all we would just need summary statistics or descriptive statistics on Behavior like if all we wanted to do was just how often does this person um make a sound again you could just count but if you wanted to say how frequently does the baby transition between happy and unhappy so in that case you need to parameterize aspects that are not directly observable that are based upon observables and that's subjective because it's about a subject so we'll talk about subjective and objective models and this is going to be introduced in terms of meta Bayesian methods which sounds very meta but once we see the figure it'll actually be revealed to be in fact the structure of Behavioral observation and this relates to many different topics we've discussed like Bayes optimality of behavior so looking at like the diversity of neural systems or cognitive systems and asking what is the generative model such that this system's behavior is Bayes optimal now that doesn't mean it's the most profitable business or you know the best free throw shooter or any of these things but still given the generative model its path of least action is Bayes optimal in some way and this relates to computational phenotyping for example in the human um domain with Psychiatry but also in other domains like ants research okay just raise your hand or write some some thoughts if you'd like so 9 2 goes into the metabasian methods the chapter deals with a utility of active inference formulations in analyzing data from behavioral experiments this goes beyond proof of principle simulations we have seen in previous chapter and exploits octave inference in answering scientific there are two related reasons for fitting a computational model to observe the first is to estimate parameters of interest from the model that best explain the behavior of a specific so one individual or a group of subjects this is computational phenotyping so computational phenotyping helps us go beyond the summary statistics of behavior and try to understand underlying differences that are potentially only accessible or are manifested through multiple types of data like fmri data EEG data behavioral data self-report data and and those kinds of computational phenotypes might be really important for forecasting behaviors or for deciding interventions so let's just imagine a

situation where a system is engaging in a behavior that we want to change so ignoring why we want to change it and all this kind of stuff um we wanted to do this and it always does that well there might be a few different reasons that that happens and we can look at this in the context of figure 4.3 our standard generative model here one option for why the system engages in the behavior is that it's a tap it so it's policy prior is that it um goes a certain way that's option one option two is that it prefers to go that direction so whether or not it's habitual for it it's expected free energy calculation sharpens that policy prior to the policy posterior and selects this action and the third option is that it has an understanding of how that action changes the world such that weather is Habitual option one and whether it's preferred option two it still believes that that course of action is the one that brings the world into alignment with preferences and those are three very different situations for example in the context of treating addiction so in the case of Habitual Behavior maybe one course of action is appropriate in the case where somebody prefers something maybe you speak directly to the preferences and then in the case where somebody's B Matrix or their causal model of the world is maladaptive then you might engage in education around how hidden States change their time so that's how you can use this type of a cognitive model to go a little bit deeper than just the phenotype manifested and and get unique explanations and predictions about systems that's the first reason and the second reason that you might want to do this kind of computational modeling is to compare alternative hypotheses that represent different explanations for a behavioral phenomenon so we're modeling ants coming in and out of the colony and we have one model it's a single parameter model there's a drive for foraging and then there's now we're going to contrast that with the two parameter model the drive for foraging and fear of predation and a three parameter model with some third proposed cause and we want to know which one of those models are helping us develop scientific explanations or effective accounts and so we can use Bayesian modeling to do that bigger 9.1 is a really interesting and uh clever diagram does anyone want to or up to this point or about figure 9.1 like give any thought or question uh Daniel I I've got a question although it's not quite formulated just yet but it's it's it had particular resonance uh to addiction because I'm in the process of uh treating such a a condition in somebody and I haven't quite formulated my question but I'll picture to you as soon as I've got it in mind we might if you don't mind go back uh if that's okay oh yeah a page or two but but I'm just still formulating it so yeah if you just Hang Ten I'll I'll get one to you soon let's look at figured I'd want to then feel free to to ask it with this setting this is kind of like just a graphical overview of the setting that was

described above and this is going to help us clarify subjective and objective modeling this is the relationship between subjective and objective models in the context of this Bayesian inference the inner dashed box so inner dashed box should look familiar it's figure 4.3 is the subjective model used by an experimental subject so that's why it's subjective is this is the the rat in the Maze this is the ants in the colony this is the person in the armchair this is who is subject to being modeling this is the subject of modeling this is the generative model that is being proposed for the subject's cognitive processes and so the same exact model that we've been talking about generative model of an agent we're just putting that into this dashed box the outer Dash box is the experimenters objective model that includes prior beliefs and and so so now let's take a step outside of just the agent of Interest outside of just the rat and the Maze now let's think about the whole Laboratory we have the scientist observing the rat in the maze which is parameterized a certain way it was it could have been 50 50 with with sugar versus bitter could have been 80 20. it could have been three flavors it could have been 22 flavors that's the parameterization of the laboratory Theta then the experimental stimuli that could be the visual stimuli that's presented or it could be the decision-making stimuli that are presented those stimuli which become observations of the subject are actions by the experimenter and analogously the actions of the subjects whether the rat goes left or right whether the person goes for the gamble or not the actions of the subject become the empirical objective observed so this is kind of like wrapping a behavioral analysis layer or an ethological layer around the generative model of an agent and so it speaks to the composability of these models this is also a little bit more of a holistic picture because it describes a setting that I believe is is non-controversial here's a person watching a bird or a scientist watching a mouse but it helps us get a bigger picture it's not just a free-floating generative it's actually the generative model especially in a laboratory or a controlled setting where the scientist is presenting stimuli that hopefully are are maximally informative with respect to differentiating this model and the outputs of that model are the inputs to our empirical data analysis so anyone have some some thoughts or or ideas on this sorry I haven't raised my hand but I can I can just uh speak as you're speaking I'm mapping myself as an objective observer in my Consulting room to a particular patient who has just brought um new um observable uh evidence in the form of a relapse and I'm just wondering how so I I hear what you you said if you went back to the if you focused on simply on the subjective model that he is using substances because it's his prefer I can't remember what those three uh things were exactly in order but it's because in the Consulting room is precisely not a control of environment uh like a mass might be in terms of

what they are exposed to and what they are able to perceive he has is being influenced by other generative models outside of the Consulting room and I think it's just Len's kind of uh um Credence to the idea of how hard it is to help or to uh help people with addictions because it's not a con in a controlled environment and there are so many generative models uh immersed or overlapping that I am not privy to and therefore only come into the picture at a particular moment when relapse has already occurred and I can't get to his brain Dynamics and inner workings of his brain directly I can only you know it's only me and him as experiencing humans at the interface rather than speaking directly to his yeah so it's just so tricky it's a very ineffective way of trying to help people I realized but help we must try yeah absolutely a lot to to um say about that um of course only for my experience as a bug doctor this is a map not a territory so so um this solid block line represents like the veil of Unknowing it's a Markov blanket ultimately so here's you and in the context of of a conversational setting you're right it's it's not a total behavioral control setting your stimuli that you are using to um appreciate influence and control or another way to think about this is the actions that you're taking which are declarative or um uh or question expressions are selected based upon expected free energy so what is expected free energy based upon well it has two components pragmatic and epistemic value right a given expression you might select your through your expertise based upon the epistemic value of that path so asking a question or conversely to um to yield pragmatic value by by by being more declarative or explicit so let's think about again those three um let's think about a generative model for that type of usage um in this setting um and and just like there's not one map for New York City it's cool there's there's not just one map but so yes definitely once this individual is out of the context now there's there's the whole ecosystem of shared intelligence that it's engaging with in an embodied way that there's so much complexity but this you're gonna um you're you're deciding which question to ask you only get one question and you're you're wanting to um again super coarsely differentiate which of three categories of patients you're dealing with right the three categories are going to be habitual but understands the world and what they prefer is healthy but their their habit is is unhealthy again let's just speak normatively this way so category one is going to be action prior is habitually addicted yes option two is there's going to be um in the C Vector so the action prior is going to be the pi option two is C preference which is over observations is such that the person prefers the outcomes like um I prefer to see um Wild colors so I'm using this thing that makes me see colors because I actually prefer the consequences of that action whether it's habitual or not so habit preference and the third option we can just coarsely say is

is um B the world model which is some learning about says that even if the person has healthy habits and healthy preferences you know it's like they don't understand the consequences of their action right now which question should you ask well without going into you know counter deception and second level um cognitive modeling of things you could ask what do you prefer or or you know category one what do you usually do category two what do you actually prefer in category three what do you think would happen if you did this so those are three distinct paths of questioning that are going to be stimuli for them which reveal an outcome action selection from them which we hope is honest which then feeds back into your model and here are the parameters might might just be the general um the context of the room and now what is the the flexibility and extensibility of this model yes what if the experimental stimuli was not just you asking a question um but what if the observation what if the stimuli was just you asking a question and the observation was what they said and EEG and fmri and biomarker and skin galvanic response yes and pupil and arousal and cicades yeah so that is why the the Sensor Fusion capacity for active inference and the composability is is really um a positive feature because we can deal with the experimental stimuli is just verbal and The observed Behavior could be this whole vector of outputs or maybe the stimuli is a whole Vector of inputs with transcranial and with this and with this and with that and the observed behavior is just self-report so whether we have single input single output multi-input multi-output we're able to make maps cognitive Maps that reflect the setting right do those always have to be discrete or distinct um effects or can that can there be a combination some sort of a uh kind of an amalgamation of um the first and the second for example of their generator model that I am choosing to do this because and you know how can they they can't be some sort of a mixture and great great question let's recall one of the um early equivalences that we brought up which is like the maximum model evidence maximizing model evidence is equivalent to minimizing surprise yes if you knew exactly how to be minimally surprised you would have maximum evidence and we can bound surprise with with variational free energy okay now model comparison rests on finding the marginal likelihoods I.E evidence for each model so we might have um three this is this is again a really simplified model but let's just say those three different um cohorts or or categories of of patients the habitual the preference and the world model so we have we have a portfolio of three of these models and then we can ask empirically given this and their behavior given the input in their behavior which one of those three models one two and three have the highest posterior evidence which one of the and so now how do you approach this as a full Bayesian consultant well the first thing you ask is

well what's the prior on these models being true let's say well that's actually 33 33 33. those are all equally likely in society or it might be known from demographic analyzes that actually the um The Habit is 90 the other two are five percent right it's just the prior on that model then we get new data and then we use that new data just like the frog jumping out of our hands we use that new data to update and re-evaluate after every question which one of that portfolio of models has what likelihood yes I'm almost thinking in this case that it makes sense that this person schedules a time to use drugs rather than try to give them up it seems possible that the best uh the the way of minimizing expected and variational free energy is to actually schedule into his generative Model A Time to use certainly that is one way to every six months uncertainty yeah under what context and and what cognitive diversity is that healthy and um exactly lion I don't know but that's the question yeah yeah fool yeah and um again this is the black line here is the markup blanket so map territory this allows us to so to project multiple and entertain multiple competing and then continually re-update that portfolio as new data come in what is the imperative for our action selection as experimenters well we want to learn that's the epistemic imperative that's the good scientist and we want to control that's the pragmatic imperative and that's the good engineer and so this is this kind of blend of like good scientists good engineer that constitutes expected free energy in a pure um epistemic setting are imperative is maximally to reduce uncertainty that's the Bayesian information maximizing principle and so we should be asking questions that are in our zone of proximal uncertainty conversely if all we cared about was pragmatic value we didn't want to learn anything didn't care to learn anything had a very fixed mindset then we would only use declarative expressions so the the this um really nicely wraps figure 4.3 again here it's just showing a single level discrete time model but this could be a hierarchical model this could be a hybrid model this could be continuous time the this subjective model the researcher proposes [Music] if the experimental stimuli don't differentiate hypotheses well they're not informative stimuli for example if we're trying to test the um efficacy of a of a cancer drug and we're testing it at um dose 1000 2000 and 3000 maybe that's all way way way too high or too low those those are not informative experiments they're not using our funding appropriately so how do we approach that well what do we do we create multiple orders of magnitude in a first experiment and then reduce our uncertainty further by pursuing data points that would give us informative values of measurements and continually update our empirical analyzes understanding the technical details is not required to use these methods effectively readers uninterested in these details are invited to skip sections nine three and nine four funny very funny now we take a

little detour into what's called variational Laplace the exact technical details of which are not super important just at the type the title level variational that means we're going to be doing the variational inference so we're going to be using energy-based methods or the evidence lower Bound in the Bayesian setting or the surprisal bounding so just everything we talked about variational inferences in play and the Laplace approximation broadly just means we're going to be using a certain kind of approximation that helps us fit a model so just a little technical subsection parametric empirical base this is really a quite applicable and important concept and it relates to um theoretical questions that people have like well if in Bayesian analysis we need to specify a prior then how do we know that our prior isn't just throwing us way off and actually we're just you know getting our um assumptions back in the analysis because we have to be explicit about our assumptions so again how do we know that we're not just getting our assumptions back after it's been laundered through the data that could be very um disconcerting so there's several approaches to this and the two major approaches to dealing with this question of Prior specification are as follows the theoretical approach here is to select priors from uh diversity of values and even different families of priors so using different families of statistical distributions poisson gaussian gamma and so on and starting from multiple different positions if you find that across families in across fine tunings of priors including weak priors and strong you find that you converge to it a single value or region that gives good evidence that like your starting point isn't that important because you converged to um in an empirically derived outcome from multiple starting points so there's enough signal in your data to recover the true value or at least the true region from from a range of starting points so that's that's one approach the empirical like Ultra empiricist approach is called parametric empirical base or PEB so we're in the classroom we're measuring children and we want to know their height we want the posterior Distribution on the children's height so again that first approach would say let's um try a gaussian of different widths centered around one two three four five six seven eight nine ten and let's try a poisson of different distributions et cetera et cetera and we converge to five feet plus or minus one parametric empirical Bays takes a really different approach that says if you have empirical data start with those empirical data and use the first measurements or all of the measurements that you get to bootstrap the prior so in that case we go into the classroom we measure 10 children their height is 5 feet plus or minus one and then we take that empirically derived descriptive statistic and utilize it as our prior so the prior is empirically generated and then we engage in standard Bayesian analysis here is a generalized linear model glm like accounts of the experimental setting um

for people who are interested in in generalized linear modeling SPM the earlier work of an ongoing of Christian at all statistical parametric modeling statistical parametric mapping is um goes into a lot more detail on this kind of notation so this section is a little bit sparse essentially the point is we can model empirical data by articulating the um the design Matrix X so that the experimental design is represented in this Matrix so if there was um you know hot and cold and uh seven days of the week you have one column for hot and cold and another column for day of the week one through seven and then each row in the experimental Matrix would be like okay um sample 12 it was hot and it was a Tuesday sample 13 it was hot and it was Wednesday sample 14 it was cold and it was a Thursday and so that Matrix the design Matrix gets separated out from the measurements uh themselves and that allows us to do this generalized linear regression approach and that can be a that can be done within this parametric empirical Bayes framework these are very short sections and uh again they were the ones that were suggested that could be skipped so there's there's a lot more to say but what we get to is this is kind of like a chapter six moment chapter six remember we had the recipe for generative model here we get the instructions for model based data analysis and again this is also very closely related to SPM here instead of starting like chapter six with the conceptualization of the system of Interest we're going to start with a behavioral ideally chapter six has already occurred before the grant was submitted or before the experiment was pre-registered or before the data were collected so that it could be done in a principled way um as they always say like I don't know the exact quote but like the statistician is called in like a forensic doctor to like consult on why the experiment went wrong but no one ever talks to the statistician before the experiment to make sure that the statistical power is um is valid there's a much pithier quote though but we start with this empirical step which is all of the data that we actually collected step two formulate um a generative model it says here a pomdp partially observable Markov decision process as in chapter 7 discrete time but it doesn't have to be in discrete time perhaps more generically we could say formulate a generative model as in chapter six that model is going to be constrained to the data that were collected not only but if you collected um temperature and humidity from a sensor you know you'd want a generative model that had the outputs of temperature and humidity several components that are required for Bayesian analysis are specified and then group level analysis occurs and figure 9.2 is going to summarize this in the context of that rat in the maze so here we go we have the data collection of the Rat and the base again Step Zero was built built the um laboratory and designed the experiment and carry it out but let's just assume that the data are just presented to

us we have several thousand um runs from several hundred mice where they were at what time maybe it's video data and we coarse grain it with with this location detection or so on but where we get to is like location Through Time we generate a plmdp with location and time as the key components that was in chapter seven and we specify a likelihood function which is the generative model message passing architecture for that generative model we as experimenters then this was steps three through six above specify likelihood here's specification likelihood function four specify the prior beliefs we had two conditions red light blue light and our prior was that uh they might have an effect of up to three standard deviations or you know up to you know such and such value but we're centering around zero or from literature we believe that um red light has a 20 increase on turning right but we're not you know too sure about that or all of and then you tilde through tilde Through Time utilda that's the empirical behavior of the Rat are observations but its actions so here's the empirical data coming in this part two three and four are kind of like the skeleton or like the the aqueduct system of our intelligence infrastructure and here comes the data flowing through this is the actual empirical results that all comes together in this model inversion step so your observations and you you are our observations the mouse's actions o are our actions its observations so we're going and Theta are all the parameters very commonly Theta is used just to mean in generalized parameters Capital Theta Theta all the parameters conditioned upon the generative model and the experiment input output is proportional to so that's what allows us to fit these models relatively parameterization given the model multiplied by factorized by separated by The observed Behavior conditioned upon Theta our stimuli and the model so here is the behavioral modeling this component is um optimizing we want our parameters ultimately and our model and our experiment itself but maybe this is already fixed and so we can't really do anything about it but we want these components conditioning upon them we want to explain the behavior I mean if we have one model that explains the mouse's Behavior 90 of the time and another model that explains it 60 of the we want that better model so that helps us parameterize the model and then that parameterization of the we can use in a generalized linear regression framework with the design Matrix and the effect sizes and a noise term so now here we have on the y-axis is um you know preference for red light or and here are the five mice that we have and that was represented in our design Matrix we had the red and the blue light and then we had the five different mics and then we're going to plot just standard statistics once we've gotten to this step so here's the empirical data step in one step two is where we make the cognitive steps three and four are kind of like further specification

of the parameterization and the cognitive model step five we it's like a particle collider or a particle inverter or we bring all the pieces together and then what comes out is a super standard group level parametric analysis and these are the kinds of figures that you would see in a scientific paper we looked at five different age cohorts and we looked at some cognitive variable again if all you wanted was um the behavioral if you want it on the y-axis just probability of going left or right well then you could just go straight from one to six you would just plot okay Mouse one through five you know here's the 50 50 line this one went 40 percent this is 75 plus or minus so if you didn't want to do cognitive you don't need steps two three five or four you just do descriptive data analysis but if you want to engage this cognitive modeling Loop with an explicitly specified cognitive in a Bayesian inference architecture here's what it looks like 96 is going to give brief examples touching upon continuous time and discrete generative models and the papers are referenced here essentially these papers model the same which is uh where metabizing inference has been successfully exploited in terms of different kinds of Behavioral data and in both cases these are icicade eye movement tasks on the left is a continuous time continuous action setting on the right is a discrete time discrete action case so continuous time model on the left discrete time model on the right some some and then again once you get to step six above after three four five you're in a total standard statistical inference setting because you're looking at linearized relationships or canonical covariates and parametric estimation at the subject level nine seven is ending the chapter with an overview on so-called false inference but there's so many aspects to this a benefit of using active inference is that IT addresses multiple dimensions of psychiatric disorders for example behavioral level psychological level and biological level I would say biological level includes behavioral and psychological and molecular but this is how they've laid it out they speak to this topic of Bayes optimality and disorder and there's a ton to say about the the socialization and the uh and pathologation of of cognitive diversity and and different people's contexts and everything like this um but just broadly working within the pathological way of thinking here are different computational pathologies that people have done mainly fristin at all have done this road map on so they've done the chapter six and they've done the chapter nine and now here are the empirical papers that bring it together provided unique explanations and predictions about these pathological settings addiction impulsivity compulsivity delusions hallucination interpersonal and personality disorders oculomotor syndromes pharmacotherapy prefrontal syndromes visual neglect disorders of interception here's a little bit uh speaking to the early categorization that we discussed with like um is somebody's Behavior due to their

habits policy prior e are they due to their preferences see are they due to their world model of how things change B are they due to their sensory perception a all of our all the variables that we know and love these are the knobs in the model this is like you make the circuit board with the nodes and the edges these are simple ones but they could be more sophisticated though that's the circuit board that's the skeleton that's the fine-tuning the model is about the the actual values quantitatively that these different nodes and edges take upon they're going to discuss a few specific interpretations and connect the computational components of these different pathologies to for example specific neurotransmitter systems neurotransmitter systems were brought up in chapter five where acetylcholine dopamine serotonin and so on were associated with certain computational um roles and so this is how we get model reification for better and worse if we find that um empirically um impulsive decision making is associated with increased dopamine that may or may not mean anything right correlation is not causation then if we find that increases in dopamine increase impulsive decision making decreases in dopamine decrease impulsive decision making all of a sudden we're getting into a more multifaceted accounts that's actually adding a lot of value and so that's how the chapter basically ends they lay out the six steps and so chapter nine is kind of like chapter six not just because the number could be rotated but because um chapter six helps us build that and then chapter nine helps us actually imbue it with data so it's an interesting chapter and it connects really really closely to the kind of empirical behavioral work that in some sense we all do because all the previous chapters it's like if you're not a theoretical Bayesian modeler those chapters seem like they're very removed from settings whereas here we're really we're in the field watching the ants we're in the room with the patient we're in the lab with the mouse and I think that makes the analysis that we do a lot more tangible anyone have any sort of thoughts or or questions or what they might want to explore next week in this who yeah well we'll we'll see where we get next week please submit questions whatever cohort wherever you're at let's add a few more questions in basic or Advanced or or related or not and um next week we'll approach the more head-on any last comments cool all right thanks Daniel thank you all see you thank you very much ciao bye